SQL ASSIGNMENT 2: BASICS, PROJECTION & FILTERING

Subject: Database Management Systems (SQL) **Topic:** DDL, DML, DISTINCT & Aliasing **Status:** Completed

TASK 1: TABLE CREATION, INSERTION & SELECTION

Create a table named `MusicPlaylist` with columns (id, song_name, artist, genre, duration). Insert 5 records representing Spotify songs, then retrieve all columns for all songs.

```
-- Create Table Structure CREATE TABLE MusicPlaylist ( id INT PRIMARY KEY, song_name VARCHAR(100), artist VARCHAR(100), genre VARCHAR(50), duration DECIMAL(4, 2) ); -- Insert 5 Records INSERT INTO MusicPlaylist (id, song_name, artist, genre, duration) VALUES (1, 'Starboy', 'The Weeknd', 'Pop', 3.50), (2, 'Blinding Lights', 'The Weeknd', 'Synthwave', 3.20), (3, 'Levitating', 'Dua Lipa', 'Pop', 3.23), (4, 'As It Was', 'Harry Styles', 'Indie Pop', 2.47), (5, 'Save Your Tears', 'The Weeknd', 'Pop', 3.35); -- Retrieve All Columns SELECT * FROM MusicPlaylist;
```

TASK 2: PROJECT COLUMNS WITH ROW LIMIT

Write a SQL query to display only `song_name` and `artist` columns from `MusicPlaylist`, showing just the first 3 records using `LIMIT`.

```
SELECT song_name, artist FROM MusicPlaylist LIMIT 3;
```

TASK 3: UNIQUE VALUES EXTRACTION

Write a SQL query on `FoodOrders` (id, restaurant, food_item, order_date) to list all unique restaurant names using `DISTINCT`.

```
SELECT DISTINCT restaurant FROM FoodOrders;
```

TASK 4: COLUMN ALIASING

Write a SQL query on `FoodOrders` to select `food_item` as 'Dish' and `order_date` as 'Date Ordered'.

```
SELECT food_item AS 'Dish', order_date AS 'Date Ordered' FROM FoodOrders;
```

TASK 5: SQL SYNTAX DEBUGGING & FIX

Fix the query: `SELECT DISTINCT food_item, restaurant FROM FoodOrders LIMIT 2` and explain the correction.

```
-- Correct & Valid SQL Syntax SELECT DISTINCT food_item, restaurant FROM FoodOrders LIMIT 2;
```

Explanation: The original query syntax is actually valid in standard SQL (ANSI/MySQL). If an error occurs in certain engines, it is due to applying `DISTINCT` improperly across multiple expressions or placing `LIMIT` before clauses like `GROUP BY` or `HAVING`. In standard SQL, `DISTINCT` evaluates unique combinations of `(food_item, restaurant)` and `LIMIT` restricts the output to 2 rows.